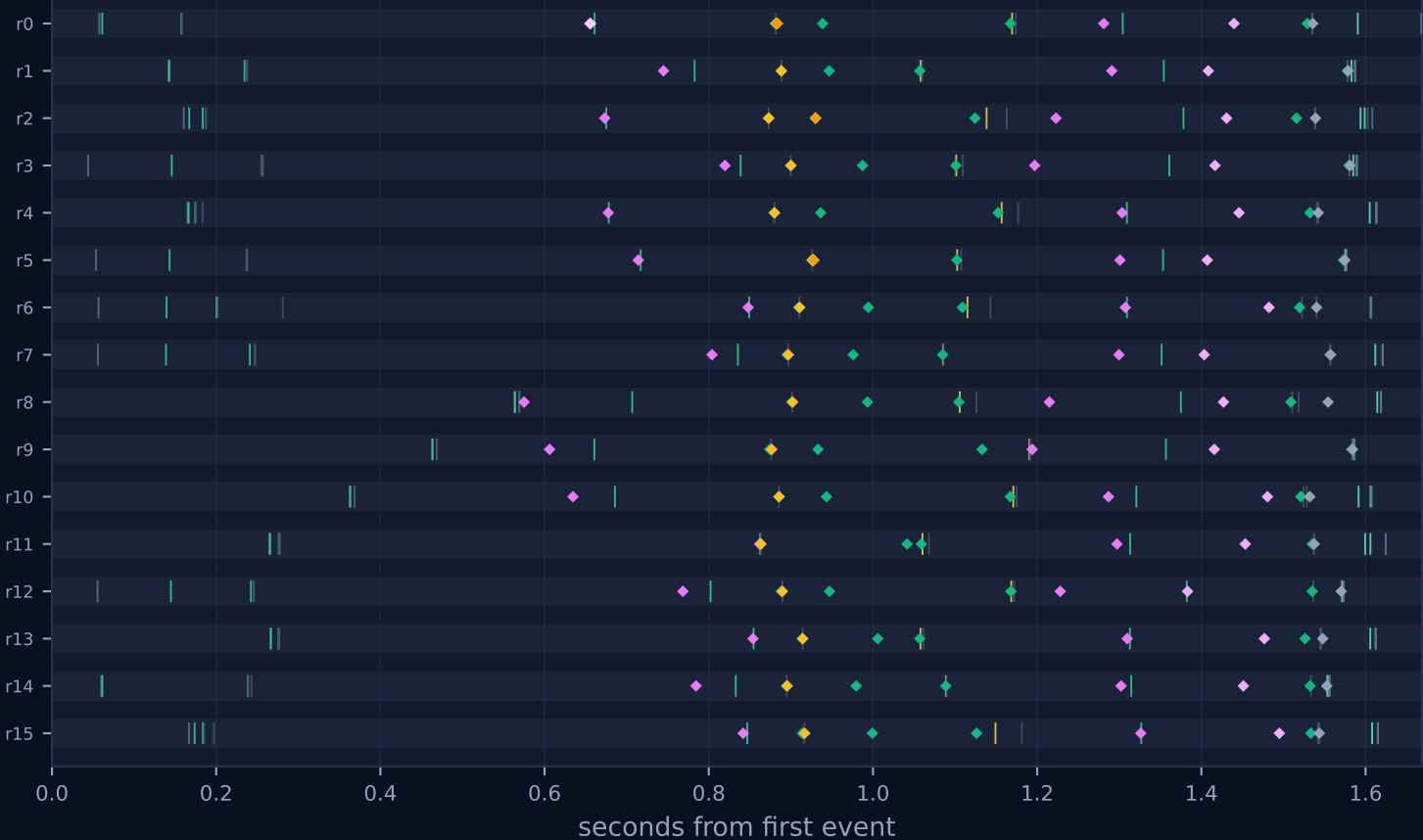




collective   window operation   reduction / arbitration   rank lifecycle